

SAMUEL CHARO MWENI

Washington, DC | +1 (202) 699 5029 | sarcharo@gmail.com | [linkedin.com-sam-mweni](https://www.linkedin.com/sam-mweni)

EDUCATION

AMERICAN UNIVERSITY

Master of Science in Data Science

Technical University of Mombasa

Bachelor of Science in Statistics and Computer Science

Washington, DC

May- 2024

Mombasa, Kenya

Nov-2019

WORK EXPERIENCE

National Institute of Health (NIH)

Research Assistant

Washington, DC

January 2024- May 2024

- Led a comprehensive evaluation of 49 text annotation tools, narrowing to 3 viable alternatives (Gate, Inception, Label Studio) for NIH's Visual Text Tool (VTT) through systematic analysis
- Developed evaluation framework using 8 critical metrics including HIPAA compliance, machine readability, and data integrity preservation
- Conducted comparative analysis focusing on ease of annotation, text originality maintenance, and machine readability, presenting findings through detailed technical documentation

Outlier

Data Science AI trainer

Maryland, MD

July 2023 – Aug2024

- Evaluated and fine-tuned Large Language Models (LLMs) using a proprietary platform, analyzing response accuracy, consistency, and alignment with training objectives
- Developed and optimized diverse training prompts across multiple domains, improving model performance and reducing bias
- Maintained 95%+ quality score in response evaluations through systematic application of evaluation guidelines and attention to detail

Cladfy Financial Services

Data Analyst

Nairobi, KE

April 2019-June2022

- Created interactive Power BI dashboards tracking 15+ financial KPIs, enabling real-time monitoring of business performance and driving 25% faster decision-making
- Developed and maintained secure BI reporting infrastructure, ensuring ODPC compliance while handling sensitive financial data for 10,000+ clients
- Conducted statistical analysis on customer transaction patterns, identifying trends that led to a 15% increase in customer retention

GRADUATE PROJECT

Political Sentiment Analysis & Voter Behavior Prediction | Python, NLP

- Developed NLP pipeline using NLTK/Transformers for 100,000+ tweets, achieving 85% sentiment accuracy
- Implemented BERT classification, TF-IDF vectorization, and statistical correlation analysis
- Visualized sentiment trends using Matplotlib/Seaborn, revealing key temporal patterns

Predictive Analysis of Police Interactions | Machine Learning

- Built predictive models (KNN, LDA, QDA) analyzing 50,000+ police records with 78% accuracy
- Performed feature engineering and model comparison using cross-validation and ROC curves
- Created Tableau visualizations for law enforcement stakeholders

SKILLS & PROFESSIONAL AFFILIATION

- **Languages & Tools:** Python (NLTK, Pandas, Scikit-learn), R, SQL, JavaScript
- **Visualization:** Tableau, Power BI, Matplotlib, Seaborn
- **Machine Learning:** Random Forest, SVM, Neural Networks, K-Means, PCA, BERT
- **Other:** Project Management, Data Governance, HIPAA/FERPA Compliance

CERTIFICATION

- Certified Data Scientist-Associate Level
- Data or Specimen Research Program